Confluence and Exact Win Probabilities for Elevens Solitaire

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Abstract

We study Elevens solitaire, a patience card game in which nine cards are laid face-up and players remove pairs summing to eleven or Jack-Queen-King triples, refilling from the deck after each removal. We prove that the win/loss outcome of any Elevens game is *confluent*: it is uniquely determined by the initial shuffle, regardless of which legal moves the player chooses. As a consequence, the win rate is a well-defined constant, computable by a rank-multiset dynamic-programming algorithm.

We determine this constant exactly:

$$P(\text{win}) = \frac{135{,}528{,}433{,}265{,}058{,}017{,}167{,}416{,}678{,}875{,}877{,}619{,}324{,}826}{1{,}267{,}970{,}709{,}742{,}491{,}186{,}578{,}322{,}321{,}454{,}365{,}697{,}265{,}625} \approx 10.689\%.$$

We also analyze a *flip variant* in which a stuck player draws the top pile card and may rescue a board match, proving the same confluence property and computing

$$P(\text{win, flip}) = \frac{8{,}148{,}818{,}506{,}107{,}124{,}832{,}421{,}641{,}049{,}420{,}090{,}762{,}431}{40{,}902{,}280{,}959{,}435{,}199{,}567{,}042{,}655{,}530{,}785{,}990{,}234{,}375} \approx 19.923\%.$$

All confluence results are kernel-verified in Lean 4; the exact win-probability computations are certified by `native_decide`, a trusted tactic that compiles Lean definitions to native OCaml code and evaluates them.

1 Introduction

Elevens is a patience (solitaire) card game. Nine cards are dealt face-up to a board; the player repeatedly removes pairs of numeric cards whose values sum to eleven (treating Ace as 1), or simultaneous triples of Jack, Queen, and King, refilling vacated slots from the deck after each removal. The game is won when all 52 cards have been removed; it is lost when no legal move remains.

Solitaire games present an interesting combinatorial challenge: given a random shuffle, what is the probability of winning? The difficulty of analyzing Klondike in particular has been described as "one of the embarrassments of applied mathematics" [10], and rigorous probabilistic analysis remains difficult for most patience variants. Elevens is unusual: we prove that *no strategy can influence the outcome*. For any fixed shuffle, either every legal sequence of play wins, or every legal sequence of play loses. We call this property *confluence*.

Confluence has two immediate consequences. First, it justifies defining the "win rate" as a single number rather than a strategy-dependent function. Second, it means the win rate can be computed by a deterministic dynamic-programming algorithm that need not consider player choices. For games without confluence — such as Klondike, where different move sequences

can lead to genuinely different game states — computing a win rate requires either a solver searching for optimal play or an assumption about player strategy. Elevens requires neither: any play yields the same outcome, so the win rate is an intrinsic property of the shuffle distribution alone.

Main results.

(i) *Outcome determinism (base game, Theorem 3.5) and Confluence (Corollary 3.7).* For any Elevens shuffle, every maximal play sequence reaches the same terminal outcome.

(ii) *Exact win probability (Theorem 4.1).* The fraction of the 52! shuffles that are winning is the rational number given in the abstract, $\approx 10.689\%$.

(iii) *Confluence and exact win probability for the flip variant (Theorems 5.2 and 5.3).* When a "flip" rescue mechanic is added, confluence holds and the win rate rises to $\approx 19.923\%$.

(iv) *Lean 4 verification.* All confluence results and both win-probability computations are formally verified; the exact rationals are certified by `native_decide`.

Related work. Most analytical attention on patience games has focused on Klondike. Its winnability is NP-complete to decide for general configurations [7], and its win rate under optimal play remains unknown despite bounding by Monte Carlo methods [10, 6] and by exhaustive solver [2]. A systematic computational study of 35 patience games [2] gives solver-determined winnability rates for many variants but does not include Elevens and does not prove confluence for any game. Constraint-programming models for Klondike are studied in [3]. Spider solitaire is similarly known to be NP-complete [9]. The most directly relevant prior work is Doyle, Grinstead, and Snell [4], who compute the exact win probability of Frustration solitaire (a simple one-player game with a single pass through the deck) using inclusion-exclusion; their paper establishes that exact rational win probabilities for patience games are both meaningful and computable. Pyramid solitaire, which shares Elevens' pairing-removal structure (pairs summing to 13), has not been studied analytically to our knowledge; no exact win probability or confluence result has been published.

Our confluence proof uses the proof structure of Newman's Lemma from abstract rewriting theory [8, 1]. Combinatorial game theory has been partially formalized in Lean 3 [5]. Our formalization is, to our knowledge, the first machine-checked result for the win probability of a patience card game.

Elevens appears not to have been studied analytically before this work. The win rate of approximately 10.69% is consistent with the authors' own Monte Carlo simulation (10^6 games: $10.68\% \pm 0.03\%$, 95% CI), but no proof of confluence or closed-form value has previously appeared.

Paper organization. Section 2 defines the game formally. Section 3 proves confluence for the base game. Section 4 presents the win-probability computation. Section 5 treats the flip variant. Section 6 summarizes the Lean 4 formalization. Section 7 discusses future work.

2 Game Setup

Definition 2.1 (Deck and shuffle)**.** *A standard deck D consists of 52 cards. Each card $c \in D$ has a* rank $\mathrm{rank}(c) \in \mathcal{R} = \{A, 2, 3, \ldots, 9, T, J, Q, K\}$ *(where T denotes the ten-card) and a suit; suits play no role in the rules. The function ν is defined only for* numeric ranks $\{A, 2, \ldots, 9, T\}$*: $\nu(A) = 1$, $\nu(T) = 10$, and $\nu(r) = r$ for $r \in \{2, \ldots, 9\}$. Face ranks $\{J, Q, K\}$ are not in the domain of ν. A* shuffle *is a bijection $\sigma : \{1, \ldots, 52\} \to D$.*

Definition 2.2 (State and rank-state)**.** *A* state *is a pair $\mathcal{S} = (B, P)$ where:*

- *$B = (b_0, \ldots, b_8)$ is the* board*, a tuple of nine slots each holding a card or $\varnothing$.*
- *$P = (p_1, p_2, \ldots)$ is the* pile*, an ordered list of undealt cards.*

The initial state *of σ is $\mathcal{S}_0(\sigma) = ((\sigma(1), \ldots, \sigma(9)), (\sigma(10), \ldots, \sigma(52)))$.*

The rank-state *is $\rho(\mathcal{S}) = (\hat{B}, \hat{P})$ where $\hat{B}$ is the multiset of ranks of non-empty board slots and $\hat{P}$ is the sequence of ranks of pile cards. The* size *of $\mathcal{S}$ is $|\mathcal{S}| = |\hat{B}| + |\hat{P}|$.*

Definition 2.3 (Legal moves and transitions)**.** *A* move *is a set of board indices $M \subseteq \{0, \ldots, 8\}$ satisfying one of:*

- Numeric pair: *$|M| = 2$, both slots non-empty, ranks r_1, r_2 with $\nu(r_1) + \nu(r_2) = 11$.*
- Face triple: *$|M| = 3$, ranks at the three slots are exactly $\{J, Q, K\}$ (one of each).*

Applying move M to state $\mathcal{S} = (B, P)$ yields $\mathcal{S}' = \mathrm{apply}(\mathcal{S}, M)$ as follows: set $b_i := \varnothing$ for each $i \in M$, then refill the vacated slots in ascending index order ($b_0 < b_1 < \cdots$) from the front of P, consuming $\min(|M|, |P|)$ cards.

A state is terminal *if it has no legal move. A terminal state is a* win *if $|\mathcal{S}| = 0$, otherwise a* loss*. Write $W(\mathcal{S}) = 1$ if $\mathcal{S}$ is winnable (some sequence of moves reaches a win).*

3 Confluence of the Base Game

The key abstraction is that moves and their outcomes depend only on the ranks of involved cards, not on suits or slot positions. We make this precise with three lemmas.

Lemma 3.1 (Rank-state equivalence)**.** *If $\rho(\mathcal{S}_1) = \rho(\mathcal{S}_2)$ then $W(\mathcal{S}_1) = W(\mathcal{S}_2)$.*

Proof. By strong induction on $|\mathcal{S}_1| = |\mathcal{S}_2|$.

Base. If $|\mathcal{S}| = 0$, both states are wins. If no legal move exists and $|\mathcal{S}| > 0$, legal-move availability depends only on $\hat{B}$, which is the same in both states; both are losses.

Inductive step. Let $|\mathcal{S}_1| = |\mathcal{S}_2| = k > 0$ with $\hat{B}_1 = \hat{B}_2$ and $\hat{P}_1 = \hat{P}_2$. If neither state has a legal move, both are losses. Otherwise both have legal moves (since legal-move availability depends only on $\hat{B}$).

Let M_1 be any legal move from $\mathcal{S}_1$, removing rank-multiset $R \subseteq \hat{B}_1$. Since $\hat{B}_2 = \hat{B}_1$ contains the same ranks with the same multiplicities, we can find a legal move M_2 from $\mathcal{S}_2$ removing the same rank-multiset R: for a numeric pair, pick any two slots in $\mathcal{S}_2$ holding ranks r and $11 - r$; for a face triple, pick any three slots holding $\{J, Q, K\}$.

After applying M_1 to $\mathcal{S}_1$ and M_2 to $\mathcal{S}_2$:

- Both boards lose the same multiset of ranks R and gain the same $k' = \min(|M|, |\hat{P}_1|) = \min(|M|, |\hat{P}_2|)$ card ranks from the front of their piles. Since $\hat{P}_1 = \hat{P}_2$, the drawn rank-prefix is the same, so $\hat{B}'_1 = \hat{B}'_2$.
- Both piles advance by k' steps, so $\hat{P}'_1 = \hat{P}'_2$.

Thus $\rho(\text{apply}(\mathcal{S}_1, M_1)) = \rho(\text{apply}(\mathcal{S}_2, M_2))$. By the inductive hypothesis (size decreases strictly), $W(\text{apply}(\mathcal{S}_1, M_1)) = W(\text{apply}(\mathcal{S}_2, M_2))$.

To conclude $W(\mathcal{S}_1) = W(\mathcal{S}_2)$, we argue both implications.

Win direction. Suppose $W(\mathcal{S}_1) = 1$, witnessed by some move M_1. The rank-identical counterpart M_2 from $\mathcal{S}_2$ satisfies $W(\text{apply}(\mathcal{S}_2, M_2)) = W(\text{apply}(\mathcal{S}_1, M_1)) = 1$ by the inductive hypothesis, so $W(\mathcal{S}_2) = 1$.

Loss direction. Suppose $W(\mathcal{S}_1) = 0$, meaning every move from $\mathcal{S}_1$ leads to a losing state. Take any move M_2 from $\mathcal{S}_2$; find the rank-identical counterpart M_1 from $\mathcal{S}_1$. By assumption $W(\text{apply}(\mathcal{S}_1, M_1)) = 0$, and by the inductive hypothesis $W(\text{apply}(\mathcal{S}_2, M_2)) = W(\text{apply}(\mathcal{S}_1, M_1)) = 0$. Since every move from $\mathcal{S}_2$ reaches a losing state, $W(\mathcal{S}_2) = 0$.

Hence $W(\mathcal{S}_1) = W(\mathcal{S}_2)$. □

Lemma 3.2 (Disjoint-move commutativity)**.** *Let M_1, M_2 be simultaneously legal moves with $M_1 \cap M_2 = \varnothing$. Then*

$$\rho(\text{apply}(\text{apply}(\mathcal{S}, M_1), M_2)) = \rho(\text{apply}(\text{apply}(\mathcal{S}, M_2), M_1)).$$

Proof. Disjointness ensures the cards at positions M_2 are untouched by M_1 (and vice versa), so M_2 remains legal after M_1.

Board rank-multiset. In either order, the $|M_1| + |M_2|$ slots in $M_1 \cup M_2$ are cleared and then refilled. The cleared multiset of ranks is the same (the original ranks at $M_1 \cup M_2$).

Pile rank-sequence. Let $k = \min(|M_1| + |M_2|, |P|)$. The refill always draws the first k cards of P in ascending-index order across the two moves combined. When M_1 is applied first, it draws $\min(|M_1|, |P|)$ cards from the front of P; M_2 then draws $\min(|M_2|, |P'|)$ cards from what remains. Together they draw exactly k cards starting from the front of P. The same holds in the other order. Hence both orderings consume the same leading rank-prefix of P and leave the same suffix.

When $|P| < |M_1| + |M_2|$, all pile cards are consumed in both orderings, and both resulting piles are empty.

In all cases the final board rank-multiset and pile rank-sequence agree, so the two rank-states are equal. □

Remark 3.3. *The two resulting boards may differ in which specific card occupies which slot, since the ascending-index refill order depends on which slots were vacated in which step. But they are rank-equal, which is all Lemma 3.1 requires.*

Lemma 3.4 (Shared-slot rank preservation)**.** *Let M_1, M_2 be simultaneously legal moves with $M_1 \cap M_2 \neq \varnothing$. Then $\rho(\text{apply}(\mathcal{S}, M_1)) = \rho(\text{apply}(\mathcal{S}, M_2))$.*

Proof. We consider the possible move-type combinations.

Both numeric pairs. $M_1 = \{i, j_1\}$, $M_2 = \{i, j_2\}$ for some shared index i. If $j_1 = j_2$ then $M_1 = M_2$ and the result is immediate. Otherwise $j_1 \neq j_2$: card b_i has rank r; since both moves are legal, b_{j_1} and b_{j_2} both have rank $11 - r$ (the unique numeric complement of r, since ν is

injective on $\{A, 2, \ldots, T\}$). Each move removes rank-multiset $\{r, 11-r\}$ from the board and draws the same two top pile cards (same pile, same count). The rank-state change is identical.

Both face triples. Each move removes $\{J, Q, K\}$ and draws the same three top pile cards. The rank-state change is identical.

One numeric pair, one face triple. Impossible: numeric ranks $\{r, 11-r\} \subseteq \{A, 2, \ldots, T\}$ and face ranks $\{J, Q, K\}$ are disjoint, so no board slot can appear in both a numeric pair and a face triple. This contradicts $M_1 \cap M_2 \neq \varnothing$. □

Theorem 3.5 (Outcome determinism)**.** *For any state $\mathcal{S}$ and any two legal moves M_1, M_2, $W(\text{apply}(\mathcal{S}, M_1)) = W(\text{apply}(\mathcal{S}, M_2))$.*

Proof. By strong induction on $|\mathcal{S}|$. The base case $|\mathcal{S}| = 0$ is vacuous (no legal moves). For the inductive step, write $\mathcal{S}_i = \text{apply}(\mathcal{S}, M_i)$.

Case $M_1 \cap M_2 = \varnothing$. By Lemma 3.2, M_2 remains legal from $\mathcal{S}_1$ and $\rho(\text{apply}(\mathcal{S}_1, M_2)) = \rho(\text{apply}(\mathcal{S}_2, M_1))$. Call these states $\mathcal{T}_1$ and $\mathcal{T}_2$. By Lemma 3.1, $W(\mathcal{T}_1) = W(\mathcal{T}_2)$.

It remains to show $W(\mathcal{S}_i) = W(\mathcal{T}_i)$. This theorem, applied inductively at size $|\mathcal{S}_i| < |\mathcal{S}|$, says any two legal moves from $\mathcal{S}_i$ lead to successors with equal W values. In particular $W(\mathcal{S}_i) = 1$ iff some successor of $\mathcal{S}_i$ has $W = 1$ iff *all* successors do (since they agree), so $W(\mathcal{S}_i) = W(\text{apply}(\mathcal{S}_i, M))$ for every legal M. Taking $M = M_{3-i}$ gives $W(\mathcal{S}_1) = W(\mathcal{T}_1)$ and $W(\mathcal{S}_2) = W(\mathcal{T}_2)$. Therefore $W(\mathcal{S}_1) = W(\mathcal{S}_2)$.

Case $M_1 \cap M_2 \neq \varnothing$. By Lemma 3.4, $\rho(\mathcal{S}_1) = \rho(\mathcal{S}_2)$, so $W(\mathcal{S}_1) = W(\mathcal{S}_2)$ by Lemma 3.1.

In both cases $W(\mathcal{S}_1) = W(\mathcal{S}_2)$, establishing that every legal move from $\mathcal{S}$ leads to the same winnability. □

Remark 3.6. *The proof structure is reminiscent of Newman's Lemma [8, 1]: the disjoint case (Lemma 3.2) plays the role of local confluence by exhibiting a common successor rank-state, while the overlapping case (Lemma 3.4) is stronger — it gives rank-state equality directly after one step, without needing a joining path. Combined with termination (each move reduces $|\mathcal{S}|$), global confluence follows. Newman's Lemma as usually stated concerns identity of normal forms; here we have outcome equivalence (win/loss), which is weaker and follows more directly.*

Corollary 3.7 (Confluence)**.** *For any shuffle σ, every maximal play sequence from $\mathcal{S}_0(\sigma)$ reaches the same terminal outcome.*

Proof. Since $W(\mathcal{S}) = 1$ iff some successor has $W = 1$, and Theorem 3.5 shows all successors agree on W, we have $W(\mathcal{S}) = W(\text{apply}(\mathcal{S}, M))$ for any legal move M — that is, W is invariant under legal moves. The game always terminates (each move reduces $|\mathcal{S}|$). Every terminal state therefore has outcome $W(\mathcal{S}_0(\sigma))$. □

Corollary 3.8. *The win rate of Elevens solitaire is a well-defined constant, independent of play strategy.*

4 Exact Win Probability

4.1 DP formulation

By confluence, the winnability of any state depends only on its rank-state. We exploit this to reduce the probability computation to a DP over rank-multiset pairs.

State representation. Index the 13 ranks as $0 = A$, $1 = 2$, $\ldots$, $8 = 9$, $9 = T$, $10 = J$, $11 = Q$, $12 = K$. Represent the board and pile each as a vector in $\mathbb{Z}^{13}_{\geq 0}$, where entry r counts cards of that rank. (We use lowercase b, m for these vectors throughout this section, reserving $\hat{B}, \hat{P}$ for the rank-state notation of Section 2.) The invariant $b_r + m_r \leq 4$ holds for each rank r throughout play.

The key observation. Confluence (Corollary 3.7) says that for any specific shuffle, the outcome is fixed regardless of play. The pile ordering is part of that shuffle: two shuffles with the same pile multiset but different pile orderings may produce different outcomes (win in one, lose in the other). The function $f(b, m)$ defined below aggregates over this: it gives the *win probability* given board vector b and pile multiset m, averaging over all uniformly random orderings of m. This is the quantity a player experiences when the board composition is known but the remaining pile is in uniformly random order.

DP recurrence. Define $f(b, m)$ as the probability of winning given board b and pile multiset m, averaging over uniformly random orderings of m. The base cases are $f(0, 0) = 1$ and $f(b, m) = 0$ if $b + m \neq 0$ and b has no legal move. When b has a legal move, choose any such move M (by Theorem 3.5 the choice does not affect f). Write $R(M) \in \mathbb{Z}^{13}_{\geq 0}$ for the rank vector of the cards removed by M: for a numeric pair removing ranks r_1, r_2, $R(M) = e_{r_1} + e_{r_2}$; for a face triple, $R(M) = e_{10} + e_{11} + e_{12}$. Let $b' = b - R(M)$ be the board after clearing the matched ranks, and let $k = \min(|M|, |m|)$ be the number of cards to draw. Then

$$f(b, m) = \frac{1}{\binom{|m|}{k}} \sum_{\substack{d \in \mathbb{Z}^{13}_{\geq 0} \\ |d| = k,\ d \leq m}} \left(\prod_{r=0}^{12} \binom{m_r}{d_r} \right) f(b' + d,\ m - d). \tag{1}$$

The weight $\frac{\prod_r \binom{m_r}{d_r}}{\binom{|m|}{k}}$ is the probability that a uniformly random size-k sample without replacement from the pile multiset m has rank composition d — the multivariate hypergeometric distribution. The sum ranges over at most $O(k^{12})$ terms but in practice only a few dozen for $k \in \{2, 3\}$.

Symmetry reduction. The five complement pairs $\{(A, T), (2, 9), (3, 8), (4, 7), (5, 6)\}$ (pairs of ranks r, s with $\nu(r) + \nu(s) = 11$) are interchangeable in the following sense: any permutation of the pair labels, or swap of the two ranks within a pair, maps legal moves to legal moves and preserves the recurrence (1). The latter holds because a legal pair move removes $\{r, s\}$ symmetrically — the move depends only on the sum $\nu(r) + \nu(s) = 11$, so relabeling $r \leftrightarrow s$ within a pair sends legal moves to legal moves and leaves the DP weights unchanged. Similarly, the three face ranks $\{J, Q, K\}$ are interchangeable (a face triple removes all three regardless of order). These symmetries preserve the initial distribution over rank-states (since the full deck has exactly 4 copies of each rank), so f is constant on symmetry orbits. The symmetry group has size $5! \times 2^5 \times 3! = 23{,}040$. Canonicalizing each state by sorting pair signatures lexicographically and choosing the smaller orientation within each pair reduces the effective state space to exactly 462,983 canonical states.

Outer average over initial deals. The initial board is a uniformly random 9-card hand from the 52-card deck; the pile is the remaining 43 cards. The probability that the board has

rank vector b (with $|b| = 9$ and $b_r \leq 4$ for all r) is

$$P(\hat{B}_0 = b) = \frac{\prod_{r=0}^{12} \binom{4}{b_r}}{\binom{52}{9}}.$$

The overall win probability is therefore

$$P(\text{win}) = \sum_{\substack{b:\, |b|=9 \\ b_r \leq 4}} \frac{\prod_r \binom{4}{b_r}}{\binom{52}{9}} \cdot f(b,\ D - b),$$

where $D = (4, 4, \ldots, 4)$ is the full deck vector.

Implementation. The DP is implemented in Python 3 using exact rational arithmetic (`fractions.Fraction`) and a canonicalization-keyed dictionary for memoization. The computation visits exactly 462,983 canonical states and completes in approximately 175 seconds on commodity hardware. The result is cross-checked using Wolfram Language.

4.2 Result

Theorem 4.1. *The exact win probability for Elevens solitaire (over uniformly random shuffles) is*

$$P(win) = \frac{135{,}528{,}433{,}265{,}058{,}017{,}167{,}416{,}678{,}875{,}877{,}619{,}324{,}826}{1{,}267{,}970{,}709{,}742{,}491{,}186{,}578{,}322{,}321{,}454{,}365{,}697{,}265{,}625} \approx 0.106886.$$

The fraction is in lowest terms. The denominator factors as $3^6 \cdot 5^9 \cdot 7^7 \cdot 11^4 \cdot 13^4 \cdot 17^3 \cdot 19^2 \cdot 23^2 \cdot 29 \cdot 31 \cdot 37 \cdot 41 \cdot 43 \cdot 47$.

This value is consistent with Monte Carlo estimates of approximately 10.689% from large-scale hobbyist simulations (10^9 games). Our own simulation (10^6 games) gives $10.68\% \pm 0.03\%$ (95% CI), consistent with the exact value.

5 The Flip Variant

5.1 Rules

Definition 5.1 (Flip transition)**.** *When a state $\mathcal{S} = (B, P)$ is terminal (no legal regular moves) and $P \neq \varnothing$, draw the top card $f = P_1$. If* $\text{rank}(f) = r$ *is a numeric rank and some board slot holds rank* $11 - r$*, remove f and one such board card, then refill the vacated slot from the new top of P. Otherwise the game is a loss. Face cards flipped are always losses; they cannot complete a sum-to-11 pair.*

The flip is *forced*: it is not a player choice. When successful, it draws two pile cards (the flipped card f and the refill) and removes two cards from play, reducing $|\mathcal{S}|$ by exactly 2 — identical in total effect to a regular pair move.

5.2 Confluence

Theorem 5.2 (Outcome determinism, flip variant)**.** *For any flip-variant state $\mathcal{S}$ and any two legal regular moves M_1, M_2, $W(\text{apply}(\mathcal{S}, M_1)) = W(\text{apply}(\mathcal{S}, M_2))$.*

Proof sketch. Lemmas 3.2 and 3.4 concern only regular moves and carry over unchanged. We extend Lemma 3.1 to cover the flip transition.

The key observation is that $\rho(\mathcal{S})$ tracks the pile as a rank *sequence* $\hat{P}$. Lemma 3.2 establishes that $\hat{P}$ advances by exactly $|M_1| + |M_2|$ steps in either order when two disjoint regular moves are applied, so the top pile card's rank is invariant to regular-move ordering.

Let $\rho(\mathcal{S}_1) = \rho(\mathcal{S}_2)$. We add two cases to the inductive proof of Lemma 3.1:

Case B_1: both states stuck, flip fails. Both have $\hat{P} = \varnothing$, or both have $\hat{P}[1]$ a face rank, or both lack a board card of rank $11 - \hat{P}[1]$ (since $\hat{B}_1 = \hat{B}_2$ and $\hat{P}_1 = \hat{P}_2$). Both lose.

Case B_2: both states stuck, flip succeeds. Both flip the same rank $r = \hat{P}[1]$ and remove the same complement rank $11 - r$ from the board; both refill from $\hat{P}[2]$. The resulting rank-states are equal, and the size decreases by 2 in both. The inductive hypothesis closes the argument.

Case C. One state has regular moves and the other does not: impossible, since regular-move availability depends only on $\hat{B}$.

With Lemma 3.1 extended, the proof of Theorem 3.5 applies verbatim. □

5.3 Exact win probability

The DP recurrence (1) is extended with a flip branch. When b has no legal move and $|m| > 0$, rank r is drawn as the flip card with probability $m_r/|m|$ (proportional to its count in the pile multiset). If $r < 10$ (a numeric rank) and the complement rank with index $9 - r$ is present on the board (i.e. $b_{9-r} > 0$; recall from Section 4 that ranks are indexed $0 = A$ through $9 = T$, so $9 - r$ is the unique index whose ν-value sums with $\nu(r)$ to 11), the flip succeeds: rank r is removed from the pile, rank $9 - r$ is removed from the board, and one slot is refilled from the remaining pile. Otherwise the game is lost. The flip recurrence is

$$f_{\text{flip}}(b, m) = \sum_{\substack{r<10 \\ m_r>0,\, b_{9-r}>0}} \frac{m_r}{|m|} \cdot \begin{cases} f_{\text{flip}}(b'',\ 0) & \text{if } |m| = 1, \\ \displaystyle\sum_{\substack{s=0 \\ m'_s>0}}^{12} \frac{m'_s}{|m| - 1}\, f_{\text{flip}}(b'' + e_s,\ m' - e_s) & \text{if } |m| \geq 2, \end{cases} \tag{2}$$

where $m' = m - e_r$, $b'' = b - e_{9-r}$ (after removing the flipped pair), and e_i denotes the i-th standard basis vector. The $|m| = 1$ case handles pile exhaustion after the flip: no refill is possible, so play continues directly from $(b'', 0)$. If the outer sum is empty, $f_{\text{flip}}(b, m) = 0$.

Theorem 5.3. *The exact win probability for the flip variant is*

$$P(\textit{win, flip}) = \frac{8{,}148{,}818{,}506{,}107{,}124{,}832{,}421{,}641{,}049{,}420{,}090{,}762{,}431}{40{,}902{,}280{,}959{,}435{,}199{,}567{,}042{,}655{,}530{,}785{,}990{,}234{,}375} \approx 0.199227.$$

The fraction is in lowest terms. The denominator factors as $3^6 \cdot 5^9 \cdot 7^7 \cdot 11^4 \cdot 13^4 \cdot 17^3 \cdot 19^2 \cdot 23^2 \cdot 29 \cdot 37 \cdot 41 \cdot 43 \cdot 47$.

Remark 5.4 (Missing factor of 31)**.** *The base-game denominator contains every prime up to 47 that divides any binomial coefficient $\binom{n}{k}$ with $n \leq 52$ and $k \in \{2, 3, 9\}$. The flip-variant denominator is identical except that 31 is absent.*

By Kummer's theorem, $31 \mid \binom{n}{2}$ *iff* $n \equiv 0$ *or* $1 \pmod{31}$*, so* $\binom{31}{2} = 31 \times 15$ *is the relevant case. Starting from pile size 43 after the initial deal, pile size 31 is reached after exactly 6 pair moves (each consuming 2 pile cards:* $43 - 12 = 31$*). Face triple moves (consuming 3 pile cards) can shift the parity and reach additional pile sizes at which 31 divides the relevant binomial coefficient. The presence of 31 in the base-game denominator and its absence from the flip-variant denominator is an observed number-theoretic fact; we leave a precise combinatorial explanation as an open question.*

6 Lean 4 Formalization

All confluence results in this paper are kernel-verified in Lean 4 with Mathlib. The exact win-probability theorems are certified by `native_decide`, which compiles Lean definitions to native OCaml and evaluates them; this constitutes a *trusted computation* (the Lean kernel checks that the definition is well-typed and that `Decidable` instances are correctly derived, but does not step through the arithmetic itself). The formalization comprises five files:

(1) `Elevens/Basic.lean`: core definitions — `NumRank` (Ace–Ten as `Fin 10`), `Rank`, `Card`, `GameState`, `Move` (`Finset (Fin 9)`), `IsLegal`, `applyMove`, `rankState`, `Winnable`, and `IsTerminal`. All other files import this one.

(2) `Elevens/Confluence.lean`: machine-checked proofs of the three lemmas and Theorem 3.5.

(3) `Elevens/ConfluenceFlip.lean`: the `WinnableFlip` inductive type with an explicit `IsTerminal` precondition on the flip constructor (needed to derive a contradiction when a regular move exists); machine-checked proof of Theorem 5.2.

(4) `Elevens/WinProb.lean`: a proof-of-concept DP for small game parameters, establishing the recurrence equation (1) via a fuel-adequacy lemma.

(5) `Elevens/WinProbFull.lean`: the full-game DP over 13 ranks with memoization via `Std.HashMap (Array ℕ) ℚ` and 23,040-fold symmetry canonicalization. The two main theorems are proved by `native_decide`:

```
theorem elevensWinRate_eq :
    elevensWinRate =
      1355284332650580171674166788758776193248 26 /
      1267970709742491186578322321454365697265625 := by
  native_decide

theorem elevensFlipWinRate_eq :
    elevensFlipWinRate =
      814881850610712483242164104942009076243 1 /
      409022809594351995670426555307859902343 75 := by
  native_decide
```

The `native_decide` tactic compiles the DP to native OCaml code and runs it inside the Lean elaborator; each proof takes approximately 42 minutes to verify (both proofs run sequentially during a single `lake build`, totalling roughly 84 minutes).

Implementation notes. Board and pile states are represented as `Array` $\mathbb{N}$ of length 13 rather than functions `Fin 13` $\to \mathbb{N}$. The function representation builds closure chains of depth proportional to game depth (≤ 26), making per-rank access $O(\text{depth})$ and causing the `native_decide` computation to become impractical. Array representation gives $O(1)$ per-rank access and reduces the wall-clock time to approximately 42 minutes per proof, during which the DP visits exactly 462,983 canonical states and performs exact rational arithmetic ($\mathbb{Q}$) throughout.

7 Conclusion and Future Work

We have proved that Elevens solitaire is confluent — player choices cannot affect the outcome — and computed its exact win probability for both the base game and a flip variant. All results are machine-verified in Lean 4. Taken together, the confluence proof and win-probability computation answer both structural and quantitative questions about this game.

Several natural extensions suggest themselves.

Pyramid solitaire. Pyramid shares Elevens' pairing-removal structure: pairs summing to 13 are removed from a triangular layout of face-up cards. Unlike Elevens, however, cards in Pyramid are arranged in overlapping rows and a card may only be removed when it is uncovered; the player therefore has genuine strategic choices about which pairs to clear first, and the game is likely not confluent. Its exact win probability appears to be an open question.

Block Ten. The immediate analogue of Elevens with target sum 10 is *Block Ten* solitaire: a 9-card board, pairs summing to 10, and same-rank pairs $\{J{+}J, Q{+}Q, K{+}K\}$ for face cards. The ten-cards cannot be removed (there is no card of rank 0 to pair with them), so each ten permanently occupies a board slot. We believe Block Ten is confluent: a ten-card arriving on the board simply reduces effective board capacity for the remainder of the game, and the same rank-state argument as the base game should apply to the remaining slots. A formal proof would require extending the rank-state equivalence lemma to account for the permanently occupied slots.

Klondike solitaire. The ultimate target in this line of research is Klondike solitaire with 3-card draw. Klondike is substantially harder: hidden information (face-down cards) makes it a partially observable Markov decision process, and the win rate is genuinely strategy-dependent. The best known bounds, $\approx 47\%$–82%, leave a large gap [2, 10, 6].

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The Lean 4 source code and Python DP script are archived at `https://github.com/shinyswellow/elevens-lean` (DOI: 10.5281/zenodo.19836615).

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